FMCG Sales Overview

Date

2014/5/23 — 2016/12/31

STATE_ID

- [ ] CA
- [ ] TX
- [ ] WI

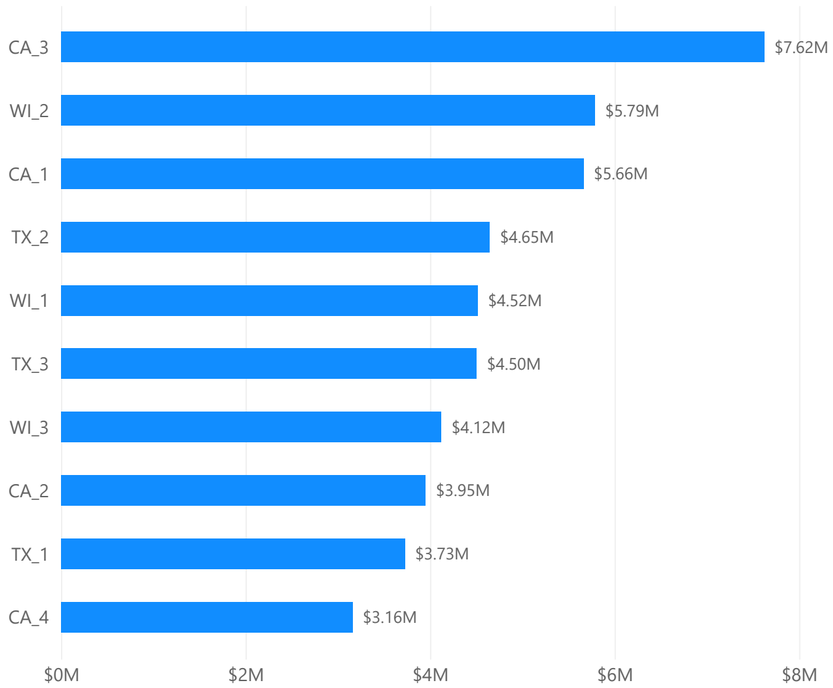
Total Revenue
$47.69M

Total Units
18.48M

Revenue per Unit
$2.58

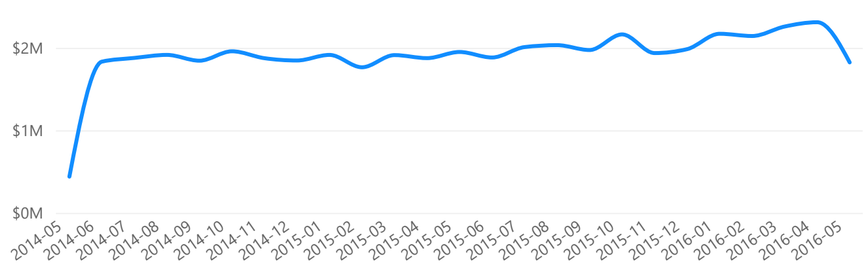
Sales Revenue by Store

Total Revenue by STORE_ID



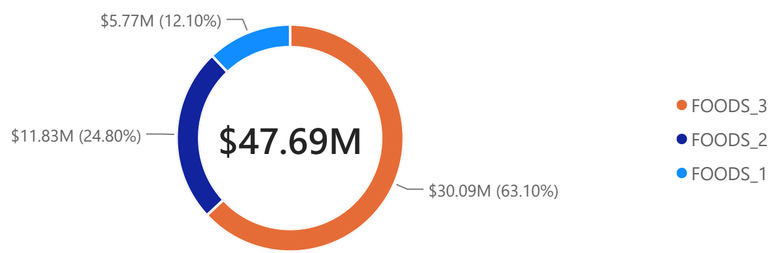
Monthly Sales Revenue

Total Revenue and Average Daily Revenue by Year Month



Sales Revenue by Department

Total Revenue by DEPT_ID



Revenue in USD. May 2014 and May 2016 contain partial-month data.

FMCG Sales Patterns

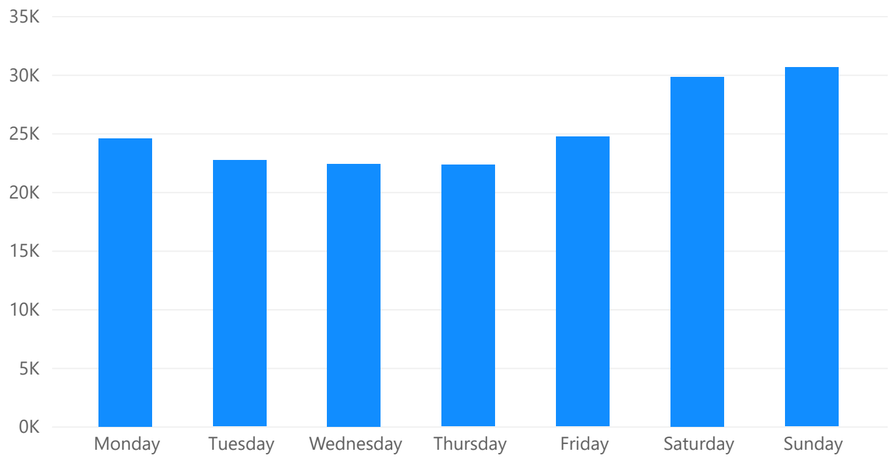
Store

STORE_ID

- [ ] CA_1
- [ ] CA_2
- [ ] CA_3
- [ ] CA_4
- [ ] TX_1
- [ ] TX_2
- [ ] TX_3
- [ ] WI_1
- [ ] WI_2
- [ ] WI_3

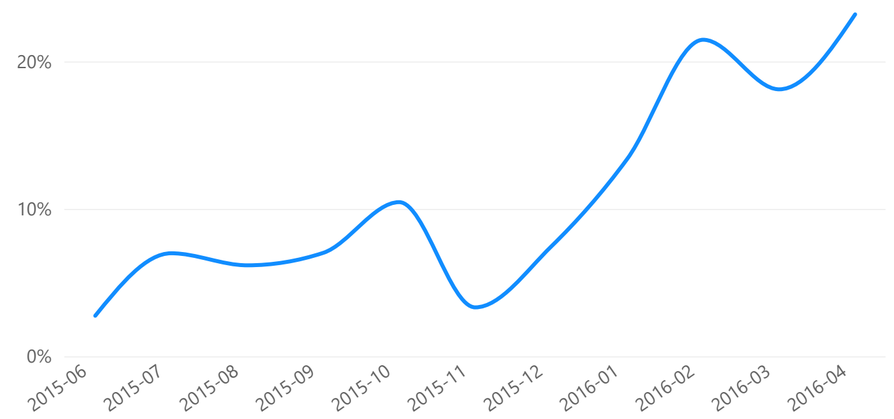
Average Daily Units by Day of Week

Average Daily Units by Day of Week



Monthly Revenue YoY Growth (Complete Months)

Monthly Revenue YoY %, Total Revenue and Revenue Last Year by Year Month

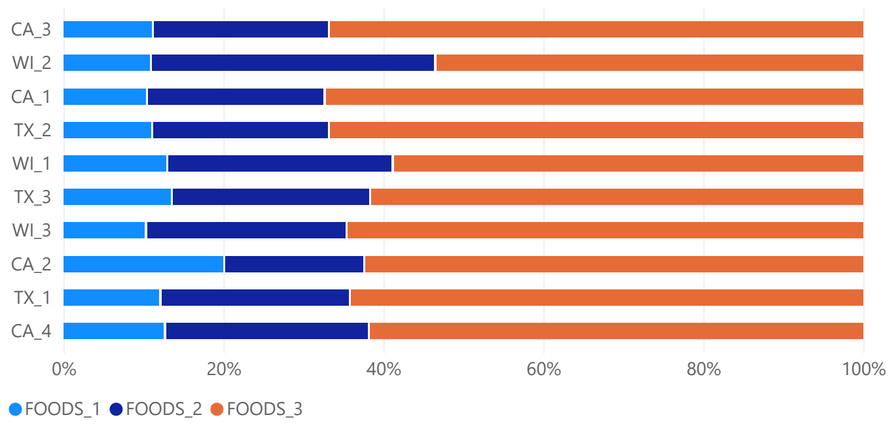


Sales Revenue by Store and Department

STORE_ID	FOODS_1	FOODS_2	FOODS_3	**Total**
CA_1	$581,656.54	$1,258,832.23	$3,822,244.74	**$5,662,733.51**
CA_2	$786,431.41	$692,164.92	$2,472,328.98	**$3,950,925.31**
CA_3	$835,868.46	$1,678,222.13	$5,105,055.26	**$7,619,145.85**
CA_4	$395,010.95	$805,972.45	$1,957,750.50	**$3,158,733.90**
TX_1	$444,121.72	$884,194.53	$2,397,503.92	**$3,725,820.17**
TX_2	$505,601.67	$1,027,399.23	$3,112,008.91	**$4,645,009.81**
TX_3	$600,921.44	$1,116,757.62	$2,785,871.12	**$4,503,550.18**
WI_1	$578,306.56	$1,273,912.88	$2,663,322.83	**$4,515,542.27**
WI_2	$622,439.40	$2,058,590.86	$3,104,458.49	**$5,785,488.75**
WI_3	$417,777.25	$1,032,538.42	$2,669,139.29	**$4,119,454.96**
Total	**$5,768,135.40**	**$11,828,585.27**	**$30,089,684.04**	**$47,686,404.71**

Department Revenue Share by Store

Total Revenue by STORE_ID and DEPT_ID



Forecast & Alerts

Alert Type

alert_type

- [ ] Above expected
- [ ] Below expected

Forecast Units — Next 28 Days

Forecast Units
841K

Historical Alerts

Alert Count
92

Historical Alert Rate

Alert Rate
10.95%

Historical WAPE — Exploratory

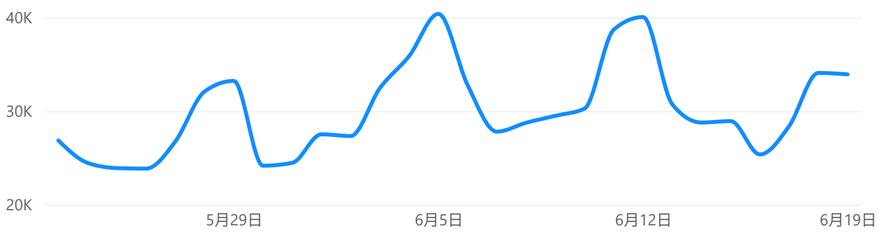
Forecast WAPE
10.38%

Historical Forecast Bias

Forecast Bias
-3.13%

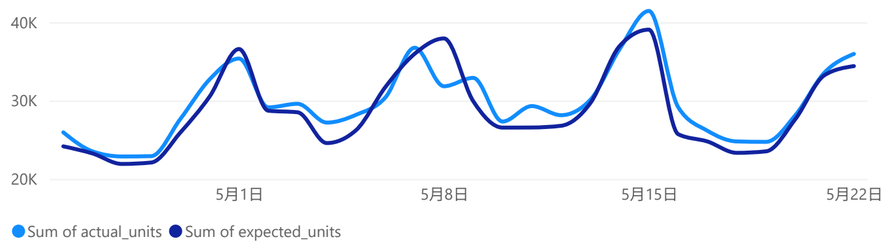
Daily Sales Forecast — Next 28 Days

Forecast Units by sales_date



Actual vs Expected Sales — Historical Review

Sum of actual_units and Sum of expected_units by sales_date

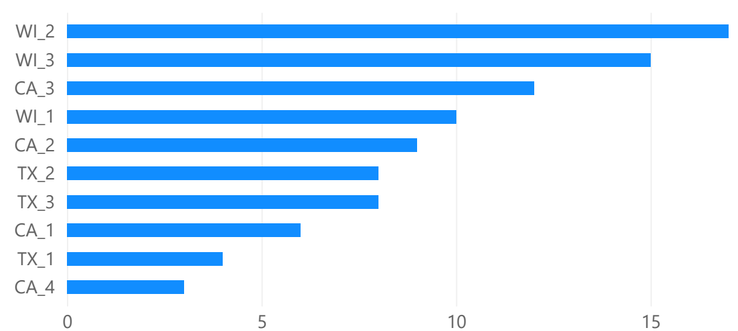


Historical Sales Alerts

sales_date	store_id	dept_id	Actual	Expected	Difference	Deviation (%)	alert_type
2016年5月9日	WI_2	FOODS_3	3388	2588	800	30.9	Above expected
2016年4月29日	TX_3	FOODS_3	1926	1541	385	25.0	Above expected
2016年5月8日	WI_1	FOODS_3	1919	2605	-686	-26.3	Below expected
2016年5月16日	WI_1	FOODS_3	1686	1312	374	28.5	Above expected
2016年4月27日	WI_1	FOODS_3	1477	1144	333	29.1	Above expected
2016年5月1日	WI_2	FOODS_2	940	1261	-321	-25.4	Below expected
2016年4月30日	WI_1	FOODS_2	871	665	206	31.0	Above expected
2016年5月21日	WI_2	FOODS_2	850	1320	-470	-35.6	Below expected
2016年5月22日	WI_2	FOODS_2	836	1196	-360	-30.1	Below expected

Historical Alerts by Store

Alert Count by store_id



Forecast: 23 May–19 Jun 2016
Historical review: 25 Apr–22 May 2016
Model: Calendar + monthly patterns
Historical evaluation is exploratory.

Alerts require both:
• ≥25% deviation from expected sales
• ≥100 units difference
Alerts flag records for investigation, not confirmed business issues.

AI-Assisted Sales Review

25 Apr–22 May 2016
• The review flagged 92 of 840 store-department-day records (10.95%): 80 above expected and 12 below expected.
• FOODS_1 accounted for 70 alerts, all above expected. Check whether the model consistently underestimates this department, including on days without alerts.
• Wisconsin stores accounted for 42 alerts (45.7%). WI_2 had the most alerts (17), followed by WI_3 (15).
• WI_2 FOODS_2 remained below expected for three consecutive days, 20–22 May, with deviations of −31.4%, −35.6% and −30.1%.

Recommended actions
Review department-level forecast bias and investigate the WI_2 FOODS_2 decline using stock availability, pricing and promotion records. Earlier analysis identified recurring monthly sales patterns; the business cause remains unconfirmed.

Limitations
Alerts identify records for investigation, not confirmed operational issues. The model upgrade and its final-period evaluation are exploratory because the features were developed after inspecting that period.